44 Esraa Ashraf Gamal

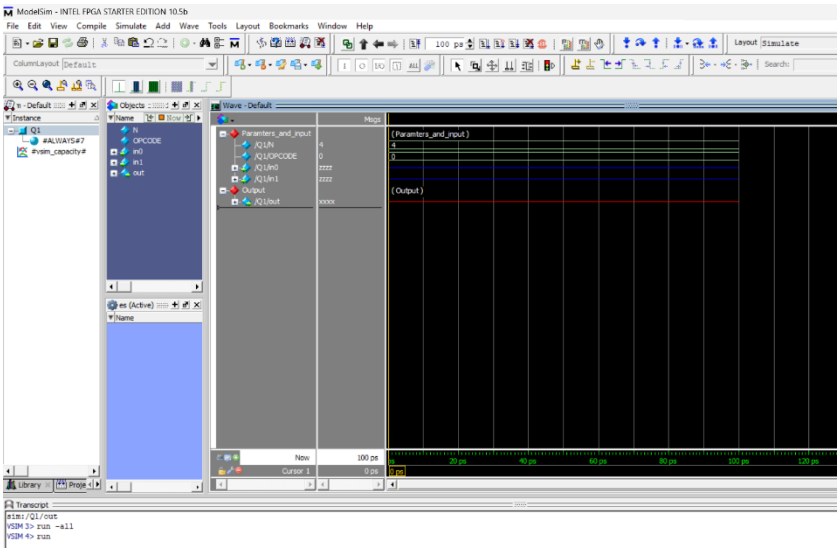
Task2

1)

the code

```
module Q1 (in0, in1, out);
    parameter N = 4;
    parameter OPCODE = 0;
    input [N-1:0] in0, in1;
    output reg [N-1:0] out;

    always @(*) begin
    case (OPCODE)
    2'b00: out = in0 + in1;
    2'b01: out = in0 | in1;
    2'b10: out = in0 - in1;
    2'b11: out = in0 ^ in1;
    endcase
    end
endmodule
```

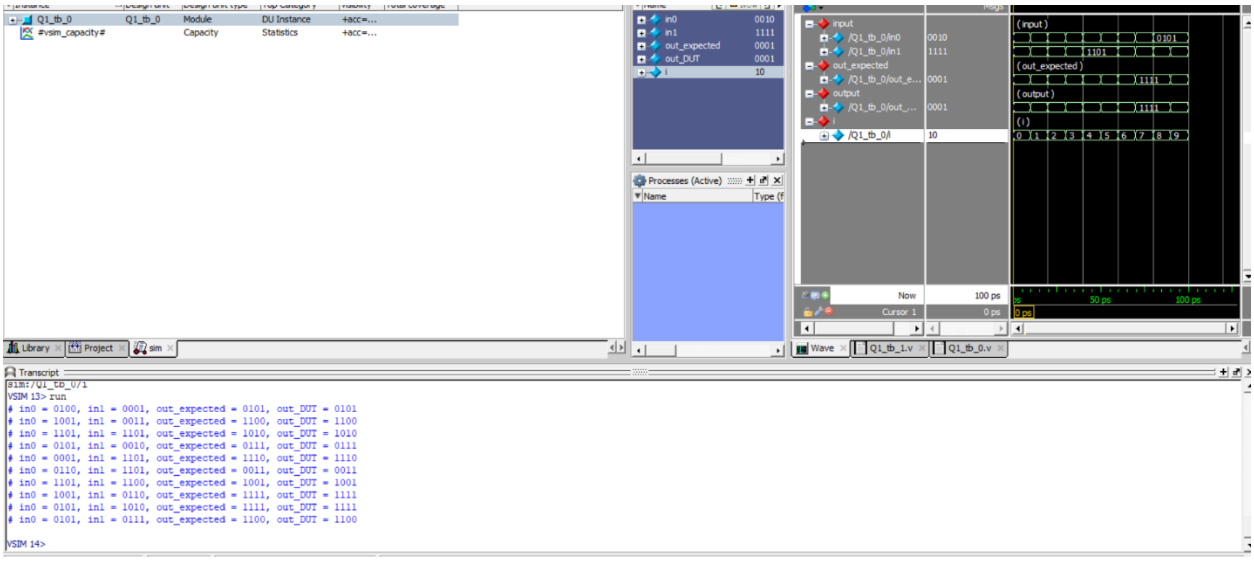


The four test bench:

1

```
module Q1_tb_0();
reg [3:0] in0, in1, out_expected;
wire [3:0] out_DUT;
Q1 #(.OPCODE(0)) DUT(in0, in1, out_DUT);
integer i;
initial begin
    for (i = 0; i < 1000; i= i+1) begin
    in0 = $random;
    in1 = $random;
    out_expected = in0 + in1;
    #10;
    if (out_expected != out_DUT) begin
    $display("Error at time %0t: in0=%b, in1=%b, expected=%b, got=%b"
            $time, in0, in1, out_expected, out_DUT);
    $stop;
    end
    end
$stop;
end

initial
$monitor("in0 = %b, in1 = %b, out_expected = %b, out_DUT = %b",
         in0, in1, out_expected, out_DUT);
endmodule
```

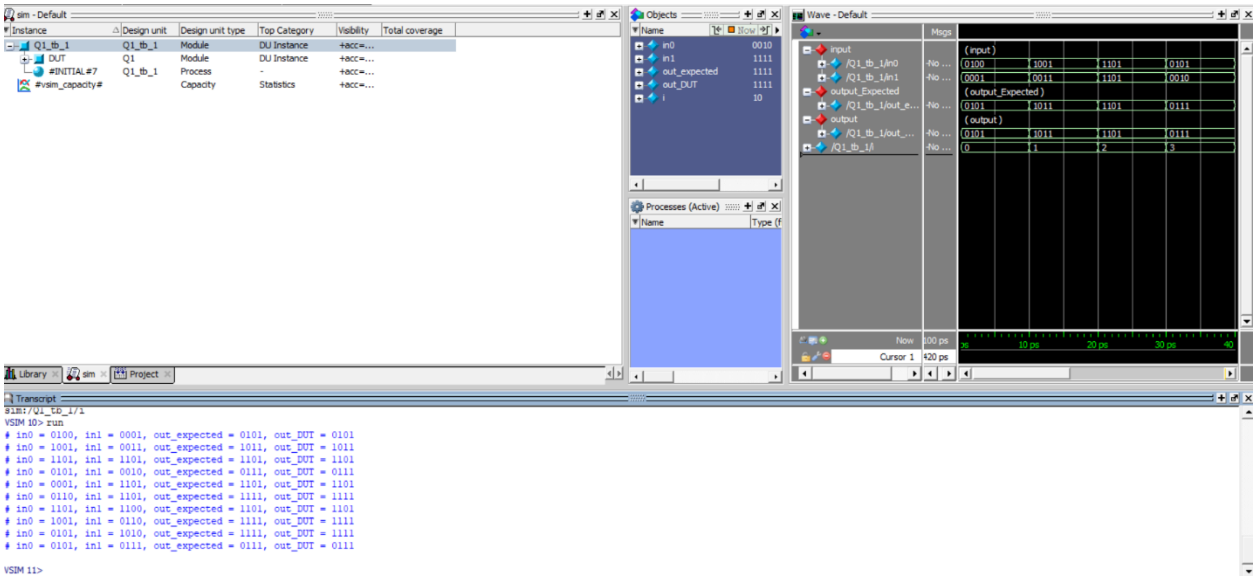


2

```
module Q1_tb_1();
reg [3:0] in0, in1, out_expected;
wire [3:0] out_DUT;
Q1 #(.OPCODE(1)) DUT(in0, in1, out_DUT);

integer i;
initial begin
    for (i = 0; i < 1000; i = i+1) begin
    in0 = $random;
    in1 = $random;
    out_expected = in0 | in1;
    #10;
    if (out_expected != out_DUT) begin
    $display("Error at time %0t: in0=%b, in1=%b, expected=%b, got=%b"
             $time, in0, in1, out_expected, out_DUT);
    $stop;
    end
    end
    $stop;
end

initial
$monitor("in0 = %b, in1 = %b, out_expected = %b, out_DUT = %b",
          in0, in1, out_expected, out_DUT);
endmodule
```

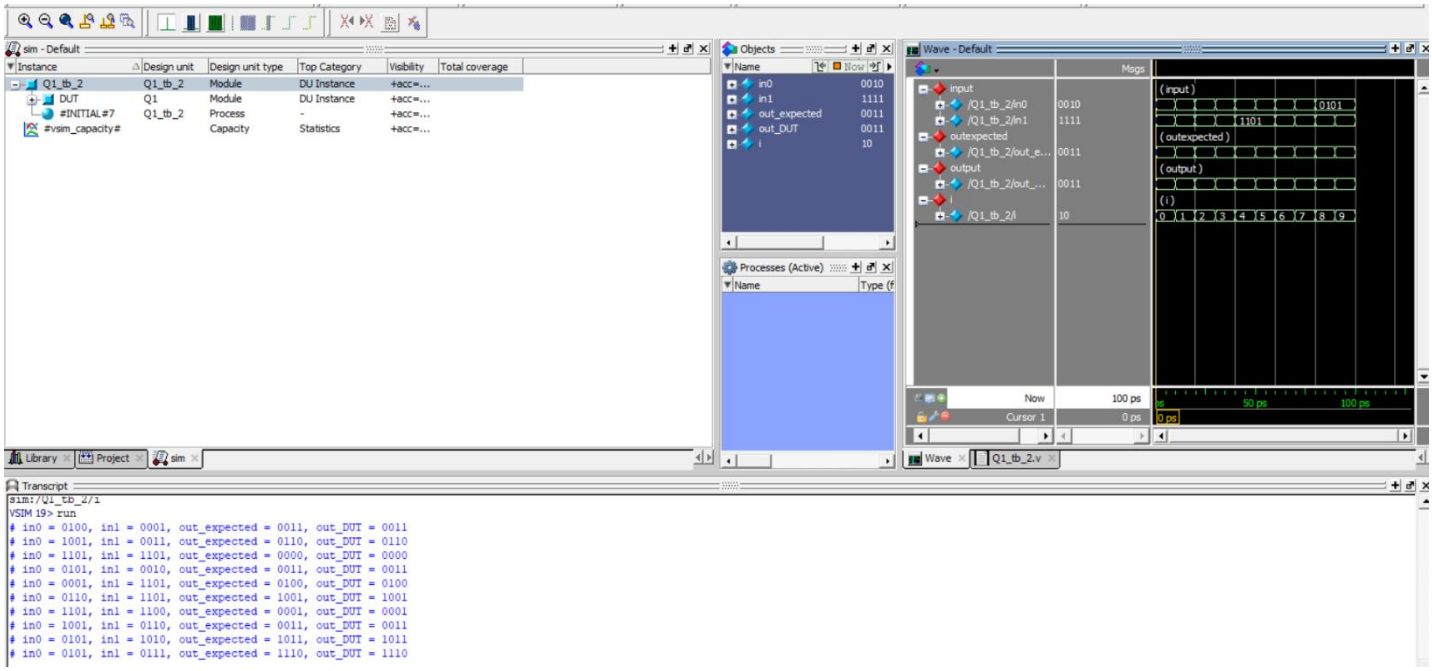


3

C:/Desktop/Diblomas/task2/Q1_tb_2.v (/Q1_tb_2) - Default

Ln#

```
1  module Q1_tb_2();
2  reg [3:0] in0, in1, out_expected;
3  wire [3:0] out_DUT;
4  Q1 #(.OPCODE(2)) DUT(in0, in1, out_DUT);
5
6  integer i;
7  initial begin
8      for (i = 0; i < 1000; i = i+1) begin
9      in0 = $random;
10     in1 = $random;
11     out_expected = in0 - in1;
12     #10;
13     if (out_expected != out_DUT) begin
14     $display("Error at time %0t: in0=%b, in1=%b, expected=%b, got=%b"
15             $time, in0, in1, out_expected, out_DUT);
16     $stop;
17     end
18     end
19     $stop;
20 end
21
22 initial
23 $monitor("in0 = %b, in1 = %b, out_expected = %b, out_DUT = %b",
24          in0, in1, out_expected, out_DUT);
25 endmodule
```



4

C:/Desktop/Diblomas/task2/Q1_tb_3.v (/Q1_tb_3) - Default

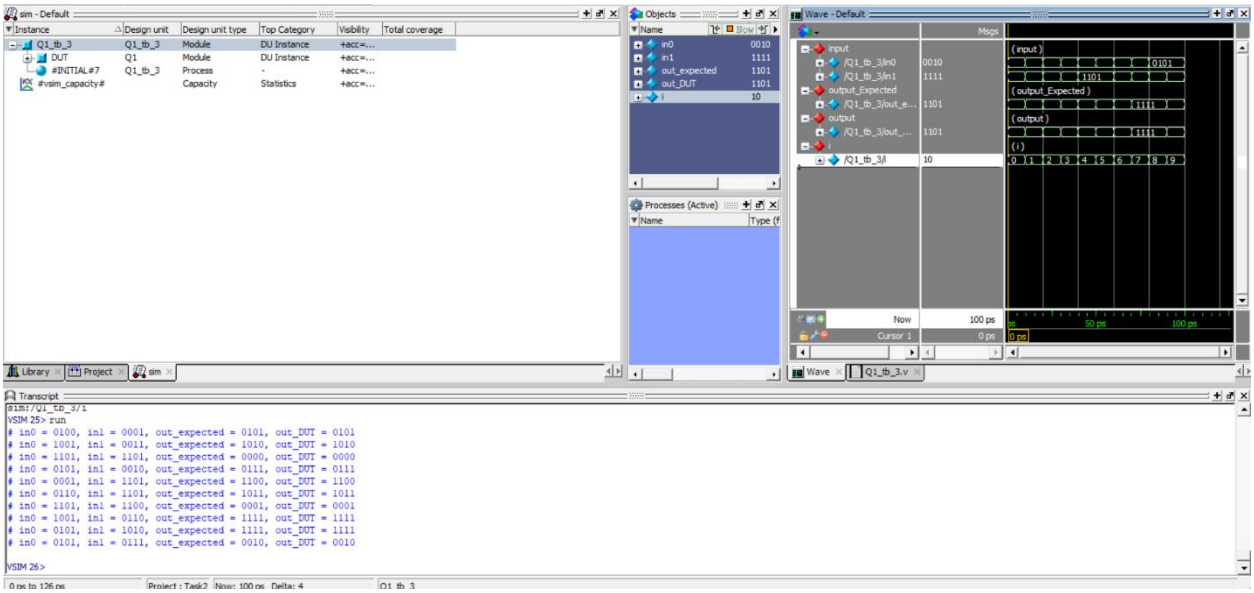
Ln#

Now

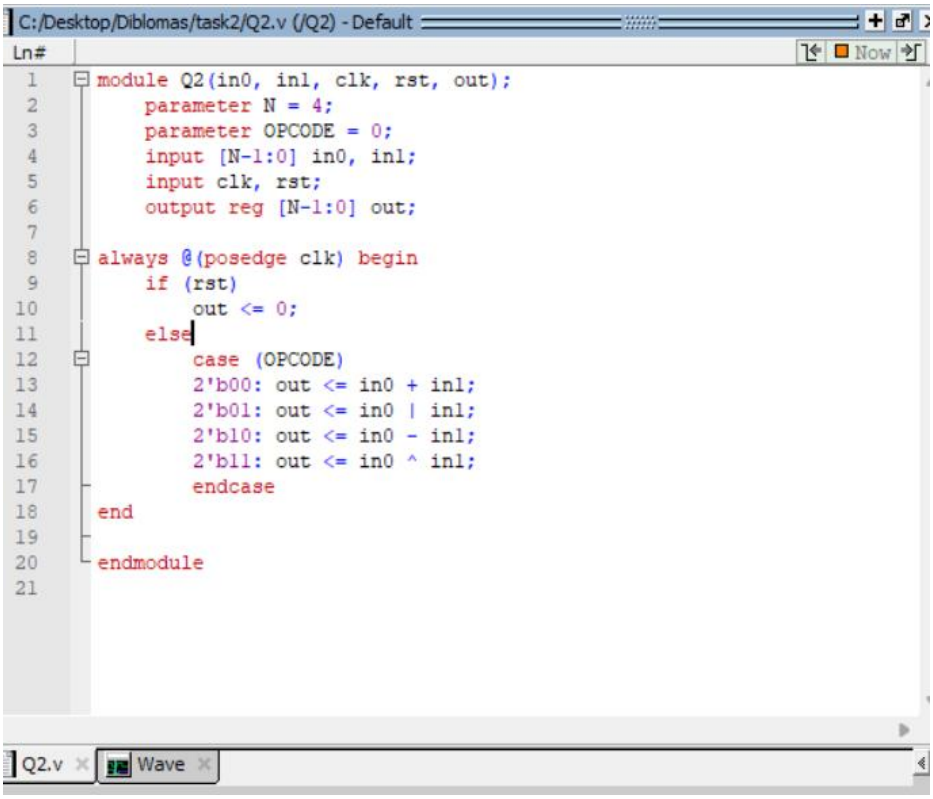
```
1  module Q1_tb_3();
2  reg [3:0] in0, in1, out_expected;
3  wire [3:0] out_DUT;
4  Q1 #(.OPCODE(3)) DUT(in0, in1, out_DUT);
5
6  integer i;
7  initial begin
8      for (i = 0; i < 1000; i = i+1) begin
9      in0 = $random;
10     in1 = $random;
11     out_expected = in0 ^ in1;
12     #10;
13     if (out_expected != out_DUT) begin
14     $display("Error at time %0t: in0=%b, in1=%b, expected=%b, got=%b"
15                $time, in0, in1, out_expected, out_DUT);
16     $stop;
17     end
18     end
19     $stop;
20 end
21
22 initial
23 $monitor("in0 = %b, in1 = %b, out_expected = %b, out_DUT = %b",
24           in0, in1, out_expected, out_DUT);
25 endmodule
```

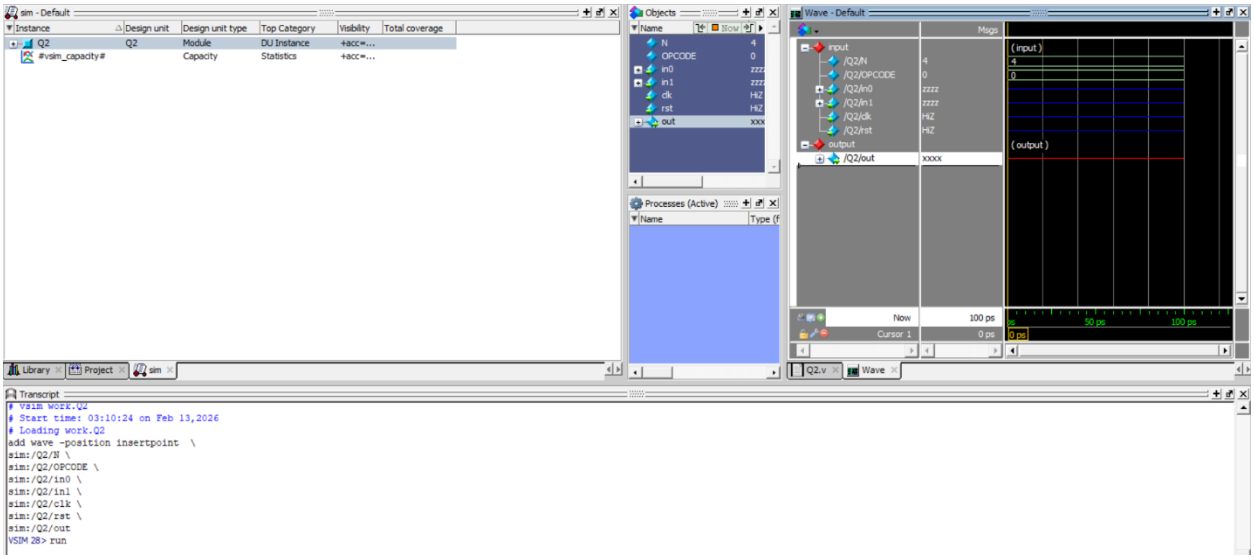
Wave

Q1_tb_3.v


sim - Default
Objects
Wave - Default
Processes (Active)
Transcript
in0 = 0100, in1 = 0001, out_expected = 0101, out_DUT = 0101
in0 = 1001, in1 = 0011, out_expected = 1010, out_DUT = 1010
in0 = 1101, in1 = 1101, out_expected = 0000, out_DUT = 0000
in0 = 0101, in1 = 0010, out_expected = 0111, out_DUT = 0111
in0 = 0001, in1 = 1101, out_expected = 1100, out_DUT = 1100
in0 = 0110, in1 = 1101, out_expected = 1011, out_DUT = 1011
in0 = 1101, in1 = 1100, out_expected = 0001, out_DUT = 0001
in0 = 1001, in1 = 0110, out_expected = 1111, out_DUT = 1111
in0 = 0101, in1 = 1010, out_expected = 1111, out_DUT = 1111
in0 = 0101, in1 = 0111, out_expected = 0010, out_DUT = 0010

2)


C:/Desktop/Diblomas/task2/Q2.v (/Q2) - Default
Ln#
1 module Q2(in0, in1, clk, rst, out);
2 parameter N = 4;
3 parameter OPCODE = 0;
4 input [N-1:0] in0, in1;
5 input clk, rst;
6 output reg [N-1:0] out;
7
8 always @(posedge clk) begin
9 if (rst)
10 out <= 0;
11 else
12 case (OPCODE)
13 2'b00: out <= in0 + in1;
14 2'b01: out <= in0 | in1;
15 2'b10: out <= in0 - in1;
16 2'b11: out <= in0 ^ in1;
17 endcase
18 end
19
20 endmodule
21
Q2.v
Wave

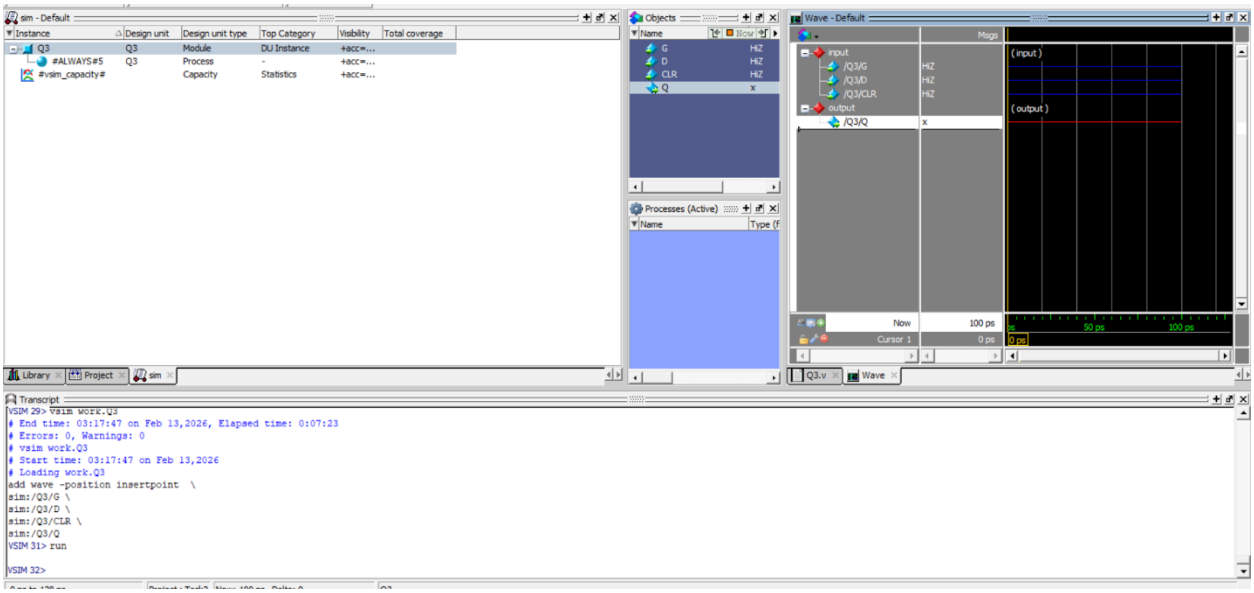


3)

```
module Q3(G, D, CLR, Q);
    input G, D, CLR;
    output reg Q;

    always @(*) begin
        if (~CLR)
            Q <= 1'b0;
        else if (G)
            Q <= D;
    end

endmodule
```

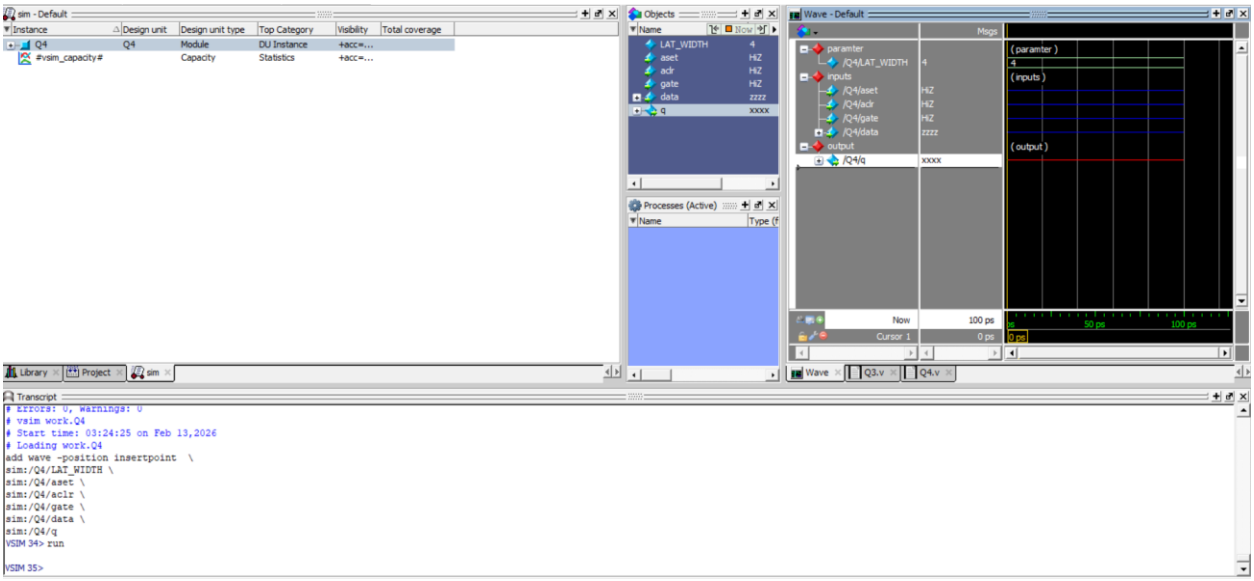

sim - Default
Instance
Design unit
Design unit type
Top Category
Visibility
Total coverage
Q3
Q3
Module
DU Instance
+acc=...
#ALWAYS#5
Q3
Process
-
+acc=...
#vsim_capacity#
Capacity
Statistics
+acc=...
Objects
Name
G
D
CLR
Q
HiZ
x
Processes (Active)
Name
Type
Wave - Default
Msgs
input
/Q3/G
/Q3/D
/Q3/CLR
output
/Q3/Q
(input)
(output)
Now
100 ps
Cursor 1
0 ps
50 ps
100 ps
Library
Project
sim
Q3.v
Wave
Transcript
VSIM 29> vsim work.Q3
End time: 03:17:47 on Feb 13,2026, Elapsed time: 0:07:23
Errors: 0, Warnings: 0
vsim work.Q3
Start time: 03:17:47 on Feb 13,2026
Loading work.Q3
add wave -position insertpoint \
sim:/Q3/G \
sim:/Q3/D \
sim:/Q3/CLR \
sim:/Q3/Q
VSIM 31> run
VSIM 32>

4)

C:/Desktop/Diblomas/task2/Q4.v - Default

```
1  module Q4 (aset, data, gate, aclr, q);
2      parameter LAT_WIDTH = 4;
3      input aset, aclr, gate;
4      input [LAT_WIDTH-1:0] data;
5      output reg [LAT_WIDTH-1:0] q;
6
7      always @(*) begin
8          if (aclr)
9              q <= 0;
10         else if (aset)
11             q <= {LAT_WIDTH{1'b1}};
12         else if (gate)
13             q <= data;
14     end
15 endmodule
16
```

Q3.v | Wave | Q4.v



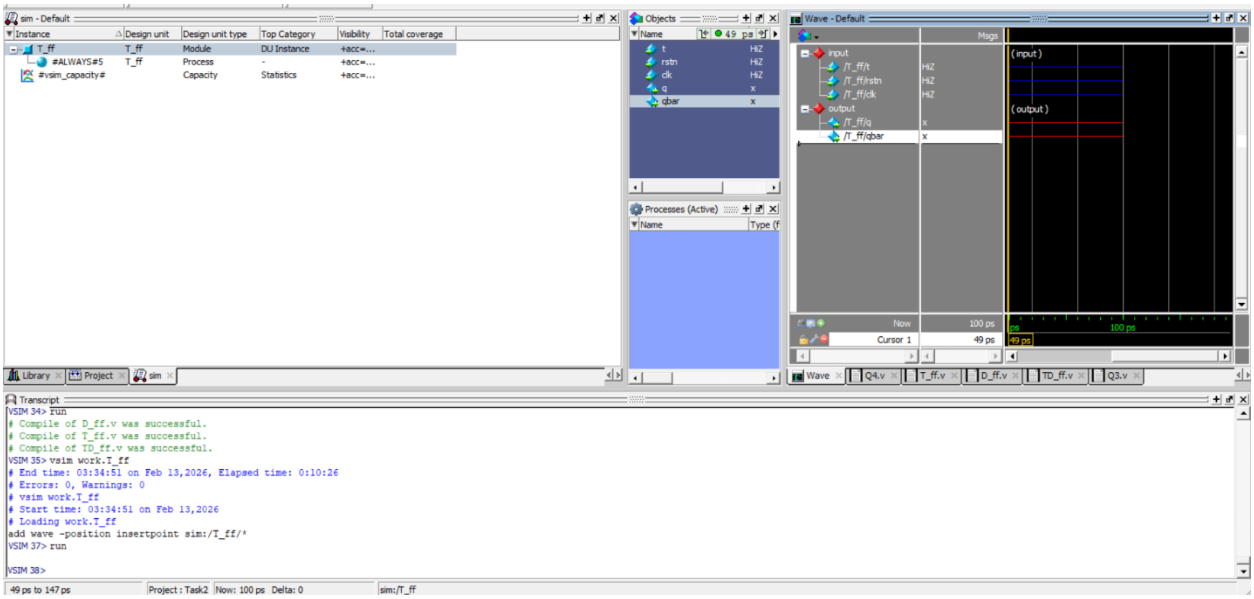
5)

a-

C:/Desktop/Diblomas/task2/T_ff.v (/T_ff) - Default

```
1  module T_ff (t, q, qbar, rstn, clk);
2      input t, rstn, clk;
3      output reg q, qbar;
4
5      always @(posedge clk or negedge rstn) begin
6          if (~rstn) begin
7              q <= 1'b0;
8              qbar <= 1'b1;
9          end
10         else if (t) begin
11             q <= ~q;
12             qbar <= ~qbar;
13         end
14     end
15 endmodule
16
```

Wave | Q4.v | T_ff.v | D_ff.v | TD_ff.v | Q3.v

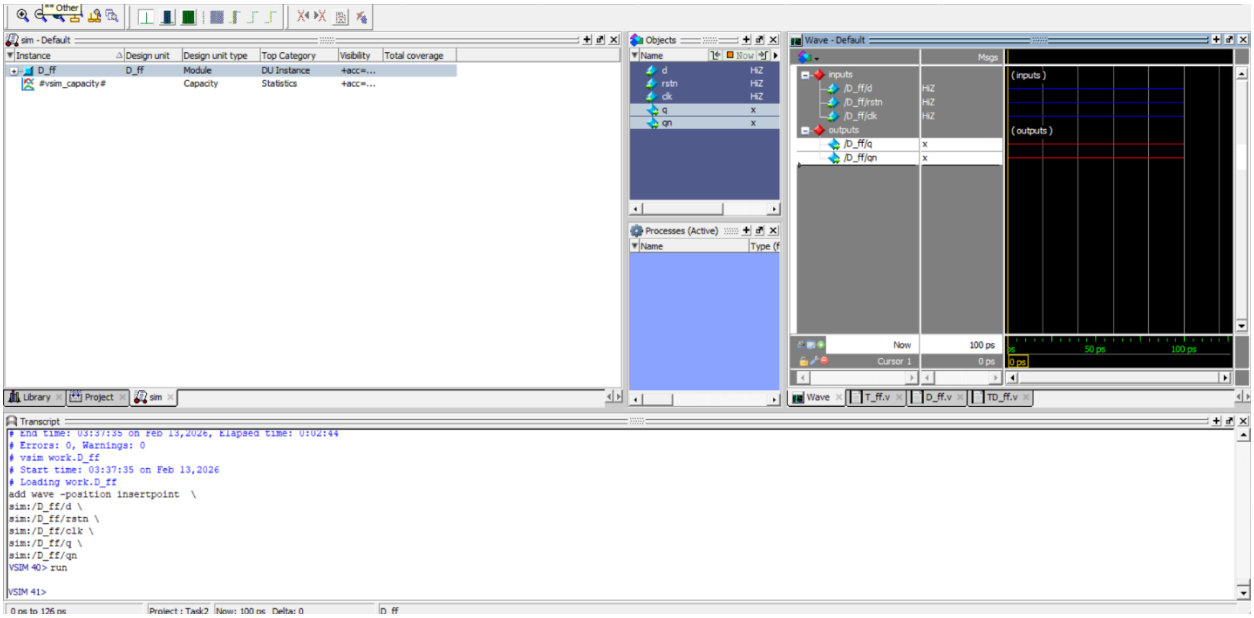


b-

C:/Desktop/Diblomas/task2/D_ff.v (/D_ff) - Default

```
1  module D_ff(d, q, qn, rstn, clk);
2      input d, rstn, clk;
3      output reg q, qn;
4
5  always @(posedge clk or negedge rstn) begin
6      if (~rstn) begin
7          q <= 1'b0;
8          qn <= 1'b1;
9      end
10     else begin
11         q <= d;
12         qn <= ~d;
13     end
14 end
15
16 endmodule
17
```

Wave | T_ff.v | D_ff.v | TD_ff.v

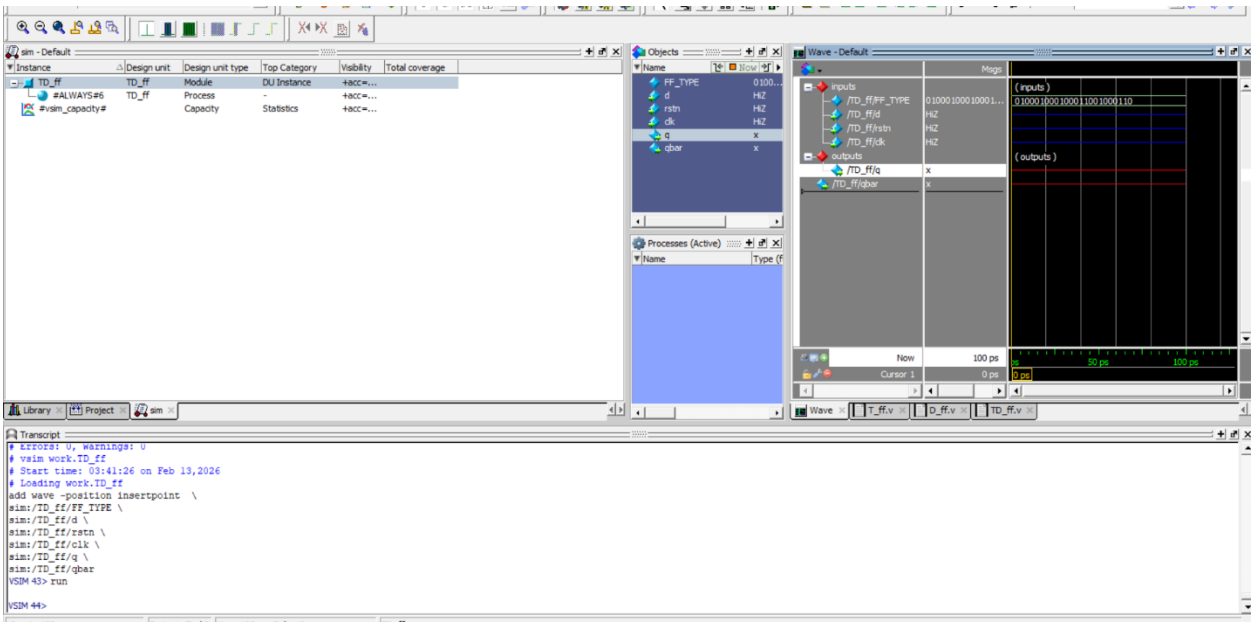


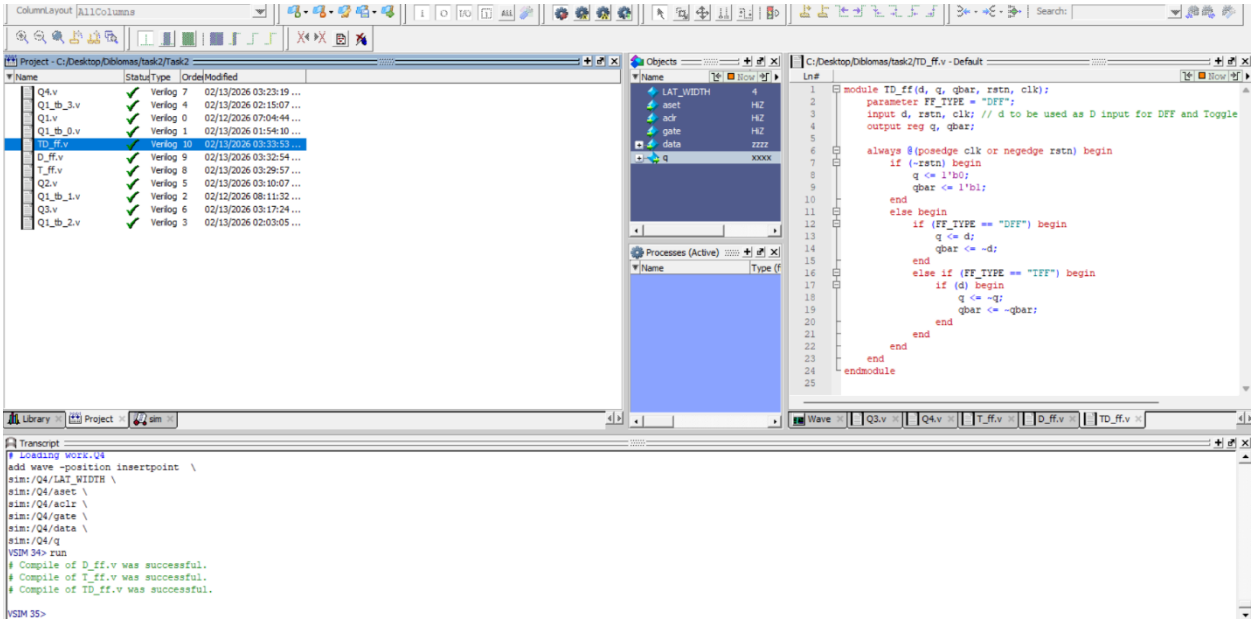
C-

C:/Desktop/Diblomas/task2/TD_ff.v - Default

```
1  module TD_ff(d, q, qbar, rstn, clk);
2      parameter FF_TYPE = "DFF";
3      input d, rstn, clk; // d to be used as D input for DFF and Toggle
4      output reg q, qbar;
5
6      always @(posedge clk or negedge rstn) begin
7          if (~rstn) begin
8              q <= 1'b0;
9              qbar <= 1'b1;
10         end
11         else begin
12             if (FF_TYPE == "DFF") begin
13                 q <= d;
14                 qbar <= ~d;
15             end
16             else if (FF_TYPE == "TFF") begin
17                 if (d) begin
18                     q <= ~q;
19                     qbar <= ~qbar;
20                 end
21             end
22         end
23     end
24 endmodule
25
```

Wave | T_ff.v | D_ff.v | TD_ff.v




Project - C:/Desktop/Diblomas/task2/Task2
Name
Statu
Type
Orde
Modified
Q4.v Verilog 7 02/13/2026 03:23:19 ...
Q1_tb_3.v Verilog 4 02/13/2026 02:15:07 ...
Q1.v Verilog 0 02/12/2026 07:04:44 ...
Q1_tb_0.v Verilog 1 02/13/2026 01:54:10 ...
TD_ff.v Verilog 10 02/13/2026 03:33:53 ...
D_ff.v Verilog 9 02/13/2026 03:32:54 ...
T_ff.v Verilog 8 02/13/2026 03:29:57 ...
Q2.v Verilog 5 02/13/2026 03:10:07 ...
Q1_tb_1.v Verilog 2 02/12/2026 08:11:32 ...
Q3.v Verilog 6 02/13/2026 03:17:24 ...
Q1_tb_2.v Verilog 3 02/13/2026 02:03:05 ...
Library
Project
sim
Objects
Name
LAT_WIDTH 4
aset HiZ
aclr HiZ
gate HiZ
data zzzz
q xxxx
Processes (Active)
C:/Desktop/Diblomas/task2/TD_ff.v - Default
Ln#
1 module TD_ff(d, q, qbar, rstn, clk);
2 parameter FF_TYPE = "DFF";
3 input d, rstn, clk; // d to be used as D input for DFF and Toggle
4 output reg q, qbar;
5
6 always @(posedge clk or negedge rstn) begin
7 if (~rstn) begin
8 q <= 1'b0;
9 qbar <= 1'b1;
10 end
11 else begin
12 if (FF_TYPE == "DFF") begin
13 q <= d;
14 qbar <= ~d;
15 end
16 else if (FF_TYPE == "TFF") begin
17 if (d) begin
18 q <= ~q;
19 qbar <= ~qbar;
20 end
21 end
22 end
23 end
24 endmodule
25
Wave
Q3.v
Q4.v
T_ff.v
D_ff.v
TD_ff.v
Transcript
Loading work.Q4
add wave -position insertpoint \
sim:/Q4/LAT_WIDTH \
sim:/Q4/aset \
sim:/Q4/aclr \
sim:/Q4/gate \
sim:/Q4/data \
sim:/Q4/q
VSIM 34> run
Compile of D_ff.v was successful.
Compile of T_ff.v was successful.
Compile of TD_ff.v was successful.
VSIM 35>